

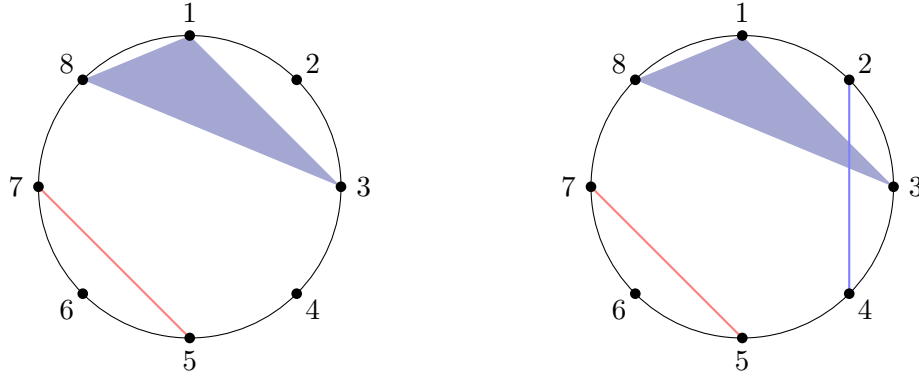
On the Structure of the Lattice of Noncrossing Partitions

Paper Summary - CHAU Dang Minh

The work focuses on properties enjoyed by the lattice of noncrossing partitions $\text{NC}(n)$, but not by the lattice of all partitions $\Pi(n)$, namely self-duality (which implies rank symmetry) and the existence of a symmetric chain decomposition. Based on these results, several combinatorial identities are derived.

1 Introduction and Notations

A *noncrossing partition* of $[n] = \{1, \dots, n\}$ is a partition of $[n]$ into blocks, such that when drawing as points in a circle in clockwise order, the convex hulls of the blocks are pairwise disjoint. For example, $\{\{1, 3, 8\}, \{2\}, \{4\}, \{5, 7\}, \{6\}\}$ is a noncrossing partition of $[8]$, while $\{\{1, 3, 8\}, \{2, 4\}, \{5, 7\}, \{6\}\}$ is not.



The *refinement order* on $\Pi(n)$ is defined by $\pi \leq \pi'$ if every block of π is contained in a block of π' . For example, $\{\{1, 3, 8\}, \{2\}, \{4\}, \{5, 7\}, \{6\}\} \leq \{\{1, 3, 8\}, \{2, 4\}, \{5, 7\}, \{6\}\}$. The *rank* of a partition π is defined as $\text{rk}(\pi) = n - \text{bk}(\pi)$, where $\text{bk}(\pi)$ is the number of blocks of π .

In the next two sections, we will show that $\text{NC}(n)$ is self-dual i.e. there exists a map $\alpha : \text{NC}(n) \rightarrow \text{NC}(n)$ such that $\alpha^2 = \text{id}$, and if $\pi \leq \pi'$, then $\alpha(\pi') \leq \alpha(\pi)$. This implies that $\text{NC}(n)$ is rank-symmetric, i.e. the number of noncrossing partitions of rank k is equal to the number of noncrossing partitions of rank $n - 1 - k$. We will also show that $\text{NC}(n)$ admits a symmetric chain decomposition, which is a partition of $\text{NC}(n)$ into chains (totally ordered subsets) such that each chain contains exactly one element of rank k for each k in some interval $[a, b]$ with $a + b = n - 1$. This implies that $\text{NC}(n)$ is rank-unimodal, i.e. the number of noncrossing partitions of rank k is less than or equal to the number of noncrossing partitions of rank $k + 1$ for all $k < \frac{n-1}{2}$, and greater than or equal to the number of noncrossing partitions of rank $k + 1$ for all $k > \frac{n-1}{2}$.